

PART - A

JAVA LAB PROGRAMS

1. Write a program to find factorial of list of number reading input as command line arguments.

```
class Factorial
{
    public static void main(String args[])
    {
        int x = args.length; // number of command line arguments
        for (int i = 0; i < x; i++) // loop through each command line argument
        {
            int n = Integer.parseInt(args[i]); // convert the argument to an integer
            if (n < 0)
            {
                System.out.println(n + " is a Negative number.");
            }
            else if (n == 0)
            {
                System.out.println("Factorial of 0 is: 1");
            }
            else
            {
                int fact = 1;
                for (int j = 1; j <= n; j++) // loop to calculate factorial
                {
                    fact = fact * j;
                }
                System.out.println("Factorial of " + n + " is: " + fact);
            }
        }
    }
}
```

Output:

```
javac Factorial.java
java Factorial 5 6 7
Factorial of 5 is: 120
Factorial of 6 is: 720
```

Factorial of 7 is: 5040

```
java Factorial 0
Factorial of 0 is: 1
```

```
java Factorial -3
-3 is a Negative number.
```

2. Write a program to display all prime numbers between two limits.

```
import java.io.*;
class Prime
{
    public static void main(String args[]) throws IOException
    {
        InputStreamReader isr = new InputStreamReader(System.in);
        BufferedReader br = new BufferedReader(isr);

        System.out.println("Enter lower limit: ");
        int lower = Integer.parseInt(br.readLine());
        System.out.println("Enter upper limit: ");
        int upper = Integer.parseInt(br.readLine());

        System.out.println("Prime numbers between " + lower + " and " + upper
+ " are: ");
        for (int i = lower; i <= upper; i++)
        {
            int count = 0;
            for (int j = 1; j <= i; j++)
            {
                if (i % j == 0)
                {
                    count++;
                }
            }
            if (count == 2)
            {
                System.out.print(i + " ");
            }
        }
    }
}
```

Output:

```
javac Prime.java
java Prime
Enter lower limit:
10
Enter upper limit:
30
Prime numbers between 10 and 30 are:
11 13 17 19 23 29
```

```
java Prime
Enter lower limit:
1
Enter upper limit:
10
Prime numbers between 1 and 20 are:
2 3 5 7
```

3. Write a program to sort list of elements in ascending and descending order and show the exception handling.

```
import java.io.*;
class Sort
{
    public static void main(String args[]) throws IOException
    {
        try
        {
            int arr[] = new int[10];
            InputStreamReader isr = new InputStreamReader(System.in);
            BufferedReader br = new BufferedReader(isr);

            System.out.println("Enter total number of elements: ");
            int n = Integer.parseInt(br.readLine());
            System.out.println("Enter " + n + " elements one by one: ");
            for (int i = 0; i < n; i++)
```

```

{
    arr[i] = Integer.parseInt(br.readLine());
}
// Add sorting logic here
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        if (arr[i] < arr[j])
        {
            int temp = arr[i];
            arr[i] = arr[j];
            arr[j] = temp;
        }
    }
}

System.out.println("Ascending order: ");
for (int i = 0; i < n; i++)
{
    System.out.print(arr[i] + " ");
}

System.out.println("\nDescending order: ");
for (int i = n - 1; i >= 0; i--)
{
    System.out.print(arr[i] + " ");
}
}
catch (ArrayIndexOutOfBoundsException e)
{
    System.out.println("The number of elements must be less than or
equal to 10 Integers.");
}
}
}

```

Output:

```

javac Sort.java
java Sort
Enter total number of elements:
5
Enter 5 elements one by one:

```

```
20
50
10
40
30
Ascending order:
10 20 30 40 50
Descending order:
50 40 30 20 10
```

4. Write a program to implement all string operations.

```
import java.io.*;
class StrOper
{
    public static void main(String args[]) throws IOException
    {
        InputStreamReader isr = new InputStreamReader(System.in);
        BufferedReader br = new BufferedReader(isr);

        System.out.println("***** STRING OPERATIONS *****");
        System.out.println("1. Concatenation of two strings");
        System.out.println("2. Comparison of two strings");
        System.out.println("3. Length of a string");
        System.out.println("4. Converting uppercase to lowercase");
        System.out.println("5. Converting lowercase to uppercase");
        System.out.println("Enter your choice: ");
        int choice = Integer.parseInt(br.readLine());

        String str1, str2, str3;
        int n;
        switch (choice)
        {
            case 1:
                System.out.println("Enter two string: ");
                str1 = br.readLine();
                str2 = br.readLine();
                str3 = str1.concat(str2);
                System.out.println("Concatenation of " + str1 + " and " +
str2 + " is: " + str3);
                break;
            case 2:
```

```

        System.out.println("Enter two string: ");
        str1 = br.readLine();
        str2 = br.readLine();
        n = str1.compareTo(str2);
        if (n == 0)
        {
            System.out.println(str1 + " and " + str2 + " are
equal.");
        }
        else if (n < 0)
        {
            System.out.println(str1 + " is less than " + str2);
        }
        else
        {
            System.out.println(str1 + " is greater than " + str2);
        }
        break;
    case 3:
        System.out.println("Enter a string: ");
        str1 = br.readLine();
        n = str1.length();
        System.out.println("Length of " + str1 + " is: " + n);
        break;
    case 4:
        System.out.println("Enter a string in uppercase: ");
        str1 = br.readLine();
        System.out.println(str1 + " in lowercase is: " +
str1.toLowerCase());
        break;
    case 5:
        System.out.println("Enter a string in lowercase: ");
        str1 = br.readLine();
        System.out.println(str1 + " in uppercase is: " +
str1.toUpperCase());
        break;
    default:
        System.out.println("Invalid choice!");
    }
}
}

```

Output:

```
javac StrOper.java
java StrOper
***** STRING OPERATIONS *****
1. Concatenation of two strings
2. Comparison of two strings
3. Length of a string
5. Converting lowercase to uppercase
Enter your choice:
1
Enter two string:
BCA
Java
Concatenation of BCA and Java is: BCAJava
```

```
java StrOper
***** STRING OPERATIONS *****
1. Concatenation of two strings
2. Comparison of two strings
3. Length of a string
4. Converting uppercase to lowercase
5. Converting lowercase to uppercase
Enter your choice:
3
Enter a string:
computer
Length of computer is: 8
```

5. Write a program to find the area of geometrical figures using method.

```
class Geometrical
{
    public int square(int side)
    {
        return side * side;
    }
    public int rectangle(int length, int breadth)
    {
        return length * breadth;
    }
    public double circle(double radius)
    {
        return 3.14 * radius * radius;
    }
    public double triangle(double base, double height)
    {
        return 0.5 * base * height;
    }
}

class Area
{
    public static void main(String args[])
    {
        Geometrical g = new Geometrical();
        double side = g.square(5);
        double areaOfRectangle = g.rectangle(4, 6);
        double areaOfCircle = g.circle(3.5);
        double areaOfTriangle = g.triangle(4, 5);
        System.out.println("Area of square: " + side);
        System.out.println("Area of rectangle: " + areaOfRectangle);
        System.out.println("Area of circle: " + areaOfCircle);
        System.out.println("Area of triangle: " + areaOfTriangle);
    }
}
```

Output:

```
javac Area.java
java Area
```

Area of square: 25.0
Area of rectangle: 24.0
Area of circle: 38.465
Area of triangle: 10.0

6. Write a program to implement constructor overloading by passing different numbers of parameters of different types.

```
import java.io.*;
class Constructor
{
    String name, regno;
    int mark1, mark2, mark3, total;
    double avg;

    Constructor() // default constructor
    {
        name = "";
        regno = "";
        mark1 = 0;
        mark2 = 0;
        mark3 = 0;
        total = 0;
        avg = 0.0;
    }

    Constructor(String name, String regno, int mark1, int mark2, int mark3)
    // parameterized constructor
    {
        this.name = name;
        this.regno = regno;
        this.mark1 = mark1;
        this.mark2 = mark2;
        this.mark3 = mark3;

        System.out.println("Student Name: " + name);
        System.out.println("Registration Number: " + regno);
        System.out.println("Marks in Subject 1: " + mark1);
        System.out.println("Marks in Subject 2: " + mark2);
        System.out.println("Marks in Subject 3: " + mark3);
    }
}
```

```

    }

    Constructor(Constructor c) // copy constructor
    {
        total = c.mark1 + c.mark2 + c.mark3;
        avg = (double) total / 3.0;
        System.out.println("Total = " + total);
        System.out.println("Average = " + avg);
    }
}

class ExConstructor
{
    public static void main(String args[]) throws IOException
    {
        InputStreamReader isr = new InputStreamReader(System.in);
        BufferedReader br = new BufferedReader(isr);

        System.out.print("Enter Student Name: ");
        String name = br.readLine();

        System.out.print("Enter Registration Number: ");
        String regno = br.readLine();

        System.out.println("Enter Marks of three Subjects: ");
        int mark1 = Integer.parseInt(br.readLine());
        int mark2 = Integer.parseInt(br.readLine());
        int mark3 = Integer.parseInt(br.readLine());

        Constructor con = new Constructor(name, regno, mark1, mark2, mark3);
        Constructor con1 = new Constructor(con);
    }
}

```

Output:

```

javac ExConstructor.java
java ExConstructor
Enter Student Name: Ram
Enter Registration Number: 16WXS85058
Enter Marks of three Subjects:
95
98
100

```

Student Name: Ram
Registration Number: 16WXS85058
Marks in Subject 1: 95
Marks in Subject 2: 98
Marks in Subject 3: 100
Total = 293
Average = 97.66666666666667

7. Write a program to calculate bonuses for different departments using method overloading.

```
abstract class Department
{
    double salary, bonus, netsalary;
    abstract void calBonus(double salary);
    abstract void display();
}

class Accounts extends Department
{
    public void calBonus(double salary)
    {
        this.salary = salary;
        bonus = salary * 0.2;
        netsalary = salary + bonus;
    }
    public void display()
    {
        System.out.println("Accounts \t" + salary + "\t" + bonus + "\t" +
netsalary);
    }
}

class Sales extends Department
{
    public void calBonus(double salary)
    {
        this.salary = salary;
        bonus = salary * 0.3;
        netsalary = salary + bonus;
    }
}
```

```

    public void display()
    {
        System.out.println("Sales \t\t" + salary + "\t" + bonus + "\t" +
netsalary);
    }
}

class Production extends Department
{
    public void calBonus(double salary)
    {
        this.salary = salary;
        bonus = salary * 0.4;
        netsalary = salary + bonus;
    }
    public void display()
    {
        System.out.println("Production \t" + salary + "\t" + bonus + "\t" +
netsalary);
    }
}

class Bonus
{
    public static void main(String args[])
    {
        double basic[] = {15000.0, 20000.0, 25000.0};
        Department dept[] = new Department[20];
        dept[0] = new Accounts();
        dept[1] = new Sales();
        dept[2] = new Production();

        for (int i = 0; i < basic.length; i++)
        {
            dept[i].calBonus(basic[i]);
        }

        System.out.println("Department \tSalary \tBonus \tNet Salary");
        for (int i = 0; i < basic.length; i++)
        {
            dept[i].display();
        }
    }
}

```

Output:

```
javac Bonus.java
```

```
java Bonus
```

Department	Salary	Bonus	Net Salary
Accounts	15000.0	3000.0	18000.0
Sales	20000.0	6000.0	26000.0
Production	25000.0	10000.0	35000.0